

E15 Calculating X_{seg} and X_{mm} from RTD data

A sample of the tracer hythane was injected as a pulse into a reactor and the exit concentration was measured as a function of time. The data is given below.

1. Construct a curve showing $C(t)$.
2. Construct the $E(t)$ curve.
3. Construct the $F(t)$ curve.
4. Compute X_{CSTR} , X_{PFR} , X_{seg} , and X_{mm} for the following elementary liquid phase reactions:
 - a. $A \rightarrow B$ with $k = 0.1 \text{ min}^{-1}$
 - b. $2A \rightarrow B$ with $k = 0.1 \text{ m}^3/\text{kmol}/\text{min}$ and $CA0 = 1.0 \text{ kmol}/\text{m}^3$

t (min)	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
C (g/m ³)	0	0.6	1.4	5	8	10	8	6	4	3	2.2	1.6	0.6	0

Decision Guide: Calculating X_{seg} and X_{MM} from RTD Data

Goal

Given RTD data:

1. Construct $C(t)$.
2. Construct $E(t)$.
3. Construct $F(t)$.
4. Calculate reactor conversion using:
 - The Segregation Model (X_{seg})
 - The Maximum Mixedness Model (X_{MM})
5. Compare these predictions for both first-order and second-order reactions.

Step 1: Start with the Experimental Data

The tracer experiment provides: Time and Tracer Concentration

Construct: $C(t)$ from the tabulated data. This is the raw experimental measurement.

Step 2: Calculate the Area Under the Curve

Compute:

$$\int_0^{\infty} C(t) dt$$

using numerical integration (typically the trapezoidal rule). This area is required to normalize the RTD.

Step 3: Construct E(t)

Calculate:

$$E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$$

Interpretation: $E(t) dt$ is the fraction of fluid that exits between t and $t + dt$.

Step 4: Construct F(t)

Calculate:

$$F(t) = \int_0^t E(t') dt'$$

Interpretation: Fraction of fluid that has exited by time t

Step 5: Determine Which RTD Model Is Being Used

Ask: Segregation Model or Maximum Mixedness Model? These models use RTD information differently.

Step 6: Solve the Segregation Model

This is the step students most frequently miss. The segregation model requires knowledge of how conversion changes with time inside a fluid element. Therefore:

1. Pretend a fluid element is a batch reactor.
2. Write the batch-reactor mole balance.
3. Solve for: $X(t)$ for the reaction of interest. Use:

$$N_{A0} \frac{dX}{dt} = -r_A V$$

CBE 3610 Reactor Design Handbook Equation (20)

Step 7: Calculate $X(t)$ for the First-Order Reaction

For:

$A \rightarrow B$

with:

$k = 0.1 \text{ min}^{-1}$

Use:

$$-r_A = kC_A$$

and solve the batch-reactor balance to obtain:

$X(t)$

Step 8: Calculate X_{seg}

The Segregation Model assumes: Each fluid element reacts independently.

The overall conversion is:

$$X_{\text{seg}} = \int_0^{\infty} X(t) E(t) dt$$

where:

- $X(t)$ = conversion that would be achieved by a fluid element after spending time t in the reactor
- $E(t)$ = exit age distribution
- X_{seg} = overall conversion predicted by the Segregation Model

X_{seg} = weighted average of all batch-reactor conversions

where the weighting is provided by the RTD function $E(t)$.

Interpretation: Weighted average of the batch-reactor conversions where the weighting comes from the RTD.

Step 9: Calculate XMM

The Maximum Mixedness Model works differently. Instead of following isolated fluid packets, it assumes fluid elements continuously mix with older fluid.

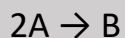
General strategy:

1. Use the previously calculated $F(t)$.
2. Use the maximum-mixedness reactor equation introduced in lecture.
3. Solve the governing differential equation.
4. Determine: X_{MM}

Because this model is built using $F(t)$, constructing $F(t)$ correctly is essential.

Step 10: Repeat for the Second-Order Reaction

Now repeat Steps 6–9 for:



using:

$$k = 0.1 \text{ m}^3/\text{kmol}/\text{min}$$

$$CA_0 = 1 \text{ kmol/m}^3$$

The batch-reactor conversion expression will be different because the kinetics are different.

Step 11: Compare the Predictions

Compare:

Xseg and XMM for:

- First-order kinetics
- Second-order kinetics

For first-order reactions: The models often give very similar results.
For nonlinear kinetics: The differences become more significant.

Useful RTD Relationships

Exit Age Distribution

$$E(t) = \frac{C(t)}{\int_0^\infty C(t) dt}$$

Cumulative Distribution Function

$$F(t) = \int_0^t E(t') dt'$$

Mean Residence Time

$$\bar{t} = \int_0^\infty t E(t) dt$$

Fraction Between Two Times

$$\text{Fraction between } t_1 \text{ and } t_2 = \int_{t_1}^{t_2} E(t) dt = F(t_2) - F(t_1)$$

Final Answer Checklist

- ✓ Constructed $C(t)$ curve.
- ✓ Calculated $E(t)$.
- ✓ Calculated $F(t)$.
- ✓ Solved the associated batch-reactor problem.
- ✓ Obtained $X(t)$ for each reaction.
- ✓ Calculated X_{seg} .
- ✓ Calculated X_{MM} .
- ✓ Compared the two RTD-model predictions.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (20) Batch Reactor Mole Balance

Eq. (44) General Rate Law

Eq. (45) Power-Law Rate Law

Key Insight

RTD Model



Requires Batch-Reactor Conversion



$X(t)$



Combine with RTD



Predict Overall Conversion